
Mechanistic Interpretability Needs Philosophy

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Abstract

Mechanistic interpretability (MI) aims to explain how neural networks work by uncovering their underlying mechanisms. As the field grows in influence, it is increasingly important to examine not just models themselves, but the assumptions, concepts and explanatory strategies implicit in MI research. We argue that *mechanistic interpretability needs philosophy* as an ongoing partner in clarifying its concepts, refining its methods, and navigating the epistemic and ethical complexities of interpreting AI systems. There is significant unrealised potential for progress in MI to be gained through deeper engagement with philosophers and philosophical frameworks. Taking three open problems from the MI literature as examples, this paper illustrates the value philosophy can add to MI research, and outlines a path toward deeper interdisciplinary dialogue.

1 Introduction

How and why do artificial neural networks produce the outputs they do? Since the resurgence of deep learning approximately a decade ago, this question has driven various efforts to interpret and explain AI systems. Among these efforts, mechanistic interpretability (MI) has emerged as an increasingly influential strand of research [Saphra and Wiegrefe, 2024]. Yet, despite its rapid rise, MI is often described as a “pre-paradigmatic” field [Bereska and Gavves, 2024]: several foundational open problems remain unsolved [Sharkey et al., 2025] and MI has faced significant critiques [Adolfi et al., 2024, Méloux et al., 2025, Makelov et al., 2024]. Making progress on these problems will require input from various perspectives and skill sets. In this paper, we focus on a cross-disciplinary contribution whose full potential has been overlooked: we argue that **mechanistic interpretability needs philosophy**—philosophy can make a crucial contribution to progress in MI. While some MI researchers have begun to acknowledge the existence of “philosophical” questions in their field [Sharkey et al., 2025, Fierro et al., 2024], the potential role of philosophy in MI remains underappreciated. MI can make greater and more efficient progress towards its scientific and societal goals through increased dialogue with philosophers and philosophical frameworks. To make our case, we examine three broad research questions in MI and show how resources from philosophy can facilitate progress on each of them. The examples we discuss serve to illustrate the various ways philosophy can add value to MI research. Our position parallels arguments that have been made for the crucial role of philosophy in the fields of physics [Rovelli, 2018], cognitive science [Thagard, 2009], economics [Nussbaum, 2016], AI research more broadly [Buckner, 2024], and science in general [Laplane et al., 2019].

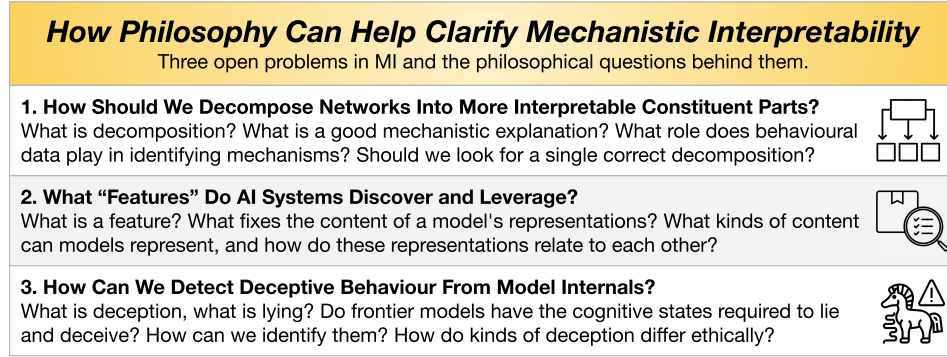


Figure 1: How philosophy can help: a case based on three open problems in MI.

What is mechanistic interpretability? Artificial neural networks are notoriously opaque. *AI Interpretability* [Ghosh and Kandasamy, 2020, Sathyan et al., 2022, Shin et al., 2022, Calderon and Reichart, 2024] and *Explainable Artificial Intelligence (XAI)* [Goebel et al., 2018, Miller, 2019, Xu et al., 2019, Dwivedi et al., 2023] encompass various approaches to address the issue of opacity.¹ While there are many XAI taxonomies [Miller, 2019, Guidotti et al., 2018, Carvalho et al., 2019, Molnar, 2020, Kotonya and Toni, 2020, Sogaard, 2021], one central distinction is between attempts to create models that are interpretable by design and attempts to interpret “black box” models [Rudin, 2019, Lakkaraju et al., 2019]. Within this latter branch, we can distinguish between the goal of generating explanations for various *non-specialist audiences* and that of developing explanations for *theorists* [Chalmers, 2025]. The latter goal requires emulating something like the scientific method, i.e., an iterative, coordinated research strategy in which a range of experimental methods are deployed, data are integrated, hypotheses are tested, and theories are refined [Kästner and Crook, 2024]. It is on this branch of the XAI tree that we find mechanistic interpretability. While there are various kinds of explanation that theorists might aim for, *mechanistic* interpretability (as the name suggests) is characterised by the search for the *mechanisms* by which a model works [Sharkey et al., 2025, Bereska and Gavves, 2024]. We discuss mechanistic explanations in detail in Section 2.1.²

Paper structure Section 2 introduces three open problems in MI and highlights specific ways that philosophy can help to make progress on those problems (see Figure 1). Section 2.1 discusses the challenge of decomposing networks into more interpretable constituent parts; Section 2.2 discusses the challenge of identifying the “features” leveraged by AI models; and Section 2.3 discusses the challenge of using MI methods to detect and mitigate deceptive behaviour by AI models. Section 3 responds to objections to our claim that MI needs philosophy and offers practical recommendations for increasing fruitful interaction between the two fields.

2 Open problems in MI – and how philosophy can help

We focus on three open problems in MI, drawing on Sharkey et al. [2025], and show how philosophy can help investigate each (see Figure 1).

2.1 How should we decompose networks into more interpretable constituent parts?

A core open problem in MI is how best to decompose complex neural networks into more interpretable constituent parts [Mueller et al., 2026, Sharkey et al., 2025]. Obvious structural components of neural networks, like neurons, parameters and attention heads, often fail to map cleanly onto functionally meaningful roles. As a result, interpretability researchers increasingly seek more abstract, distributed,

¹We will use these terms interchangeably, though we note that others use “interpretability” and “explainability” to delineate two different research agendas [Miller, 2019].

²While “mechanistic interpretability” is sometimes used narrowly to refer to a specific, culturally identified research community [Saphra and Wiegrefe, 2024] (e.g. to those associated with Distill.pub’s *Circuits* thread [Olah et al., 2020] and Anthropic’s *Transformer Circuits* thread [Elhage et al., 2022, Olsson et al., 2022]), we adopt a broader usage that includes (e.g.) prior and parallel work in the NLP and computer vision communities.

or coarse-grained decompositions that better capture a model’s internal logic and behaviour [Bricken et al., 2023, Shafran et al., 2025]. While this is often framed as a purely empirical task, empirical work may be guided by unexamined assumptions about the criteria for a good explanation. Here, philosophy of science can offer useful frameworks and insights. As others have argued [Kästner and Crook, 2024], the field of MI ultimately seeks what philosophers call *mechanistic explanations*—accounts of how phenomena arise from organized causal interactions among parts [Machamer et al., 2000, Craver, 2007, Bechtel and Abrahamsen, 2005].³ Common across fields like biology, physics, cognitive science, and economics, explanations of this sort identify a *mechanism*: a set of *entities* and *activities*, *organised* to produce or maintain a phenomenon [Glennan et al., 2022]. Mechanistic explanations do not just describe regularities but show *how* they emerge from causal structure.⁴ Thus, a key virtue of such explanations is that they support *intervention*: by revealing the components and activities responsible for a phenomenon, they clarify how it might be changed or controlled [Woodward, 2005, Craver and Darden, 2013]. The philosophical literature on mechanistic explanation provides a number of insights that can guide attempts to decompose networks into interpretable components, two of which we highlight below.

Emphasising the interrelation of mechanism and behaviour Mechanistic interpretability is often presented as targeting the “inner” workings of neural networks, in contrast to approaches that focus on input-output behaviour [Räuker et al., 2023, Vilas et al., 2024, Grzankowski, 2024]. Indeed, a common motivation for MI is the idea that, for any given task, there are many possible algorithms or solutions [see, e.g. Zhong et al., 2023], and thus a model’s ability to achieve some input–output function does not tell us *how* the function is carried out. It is thus tempting to dismiss “behavioural” approaches to studying AI systems as being distinct from (and perhaps inferior to) “mechanistic” approaches. However, from the perspective of philosophical accounts of mechanistic explanation, this contrast between mechanistic and behavioural approaches is, at best, misleading.

Behaviour is one important source of data that can inform and constrain hypotheses about algorithms. In the philosophy of neuroscience, several theorists have stressed the indispensability of behavioural data in formulating plausible models, or suggesting “sketches” for mechanisms [Krakauer et al., 2017, Piccinini and Craver, 2011]. Philosophers Budding and Zednik [2024] point out that the same approach can fruitfully extend to mechanistic interpretability. Studying behaviour should not be construed narrowly as benchmarking a model’s success at a task. A more fruitful approach, relevant to the explanatory goals of MI, involves carefully designed studies that systematically map unexpected behaviours in edge cases, identify patterns of breakdown, and test for other behavioural “signatures” of specific algorithms [Taylor et al., 2022].⁵ These insights from the philosophy of science can help refocus the efforts of MI, ensuring the field does not overlook the many ways behavioural data might inform decompositions of neural networks.

Challenging the assumption of the One True Decomposition Discussions of decomposition in MI are often framed as the search for the *right* level of analysis, implying a privileged cut through a network that reveals its “true” structure. Drawing on a Platonic metaphor, Sharkey et al. [2025] describe this aspiration as “carving neural networks at their joints” (p. 13). But this metaphor, while evocative, imports a problematic assumption: that complex systems have a unique and natural decomposition, independent of explanatory context.

Philosophers of science have long emphasized that mechanisms span multiple levels of organization [Craver, 2015], and mechanisms in biology, neuroscience, and cognitive science rarely submit to a single, correct level of analysis. In the life sciences, scientists might study whole ecosystems, individual organisms, systems of organs, mechanisms within cells, or molecular interactions. One view of how these perspectives fit together is in terms of mechanisms nested within mechanisms—what we treat as a simple activity of a component at one level (e.g. a neuron firing) can be subjected to a further “how does it work?” question, which can often be answered in terms of a lower-level mechanism (e.g. opening and closing ion channels). Importantly, no single level of description has a unique claim to being mechanistic—there are simply different levels of mechanisms. And taking a

³Older work in XAI has had similar aspirations [Balkir et al., 2022]; we happily extend the label of MI to include this work, too.

⁴Thus, philosophers sometimes contrast mechanistic models with merely *descriptive* (or “phenomenal”) models, which don’t aim to capture the causal structure of a system [Kaplan and Craver, 2011].

⁵For examples, see some recent work on Theory of Mind [Ullman, 2023, Strachan et al., 2024] and reasoning [Nezhurina et al., 2025, Lewis and Mitchell, 2025] capacities in LLMs.

mechanistic approach need not involve treating lower-level mechanisms as more important (or more “real”) than higher-level ones.⁶

In practice, which level of mechanism is most important depends largely on the pragmatic goals of researchers. This insight has direct implications for MI: different decompositions may be more or less useful, whether we are trying to (say) control outputs, understand generalisation, or detect deception. One philosophical framework for thinking about this is *explanatory pluralism*—the view that scientific understanding often requires integrating multiple, cross-cutting models, tailored to different explanatory aims [Mitchell, 2023].⁷ Taken together, these philosophical insights can help to reframe and guide efforts to decompose neural networks into components. Decompositions should be evaluated not by how well they mirror the unique “real” structure of the network, but by how effectively they support causal understanding, prediction, and intervention across different research contexts.⁸

2.2 What “features” do AI systems discover and leverage?

While there are many kinds of natural and human-engineered mechanisms, a core, if not universal, working assumption in MI is that the fundamental components of deep neural networks are *features*. Unpacking, clarifying, and interrogating this assumption are further ways philosophers can make a practical contribution to MI.

At a first pass, features involve a correspondence between an aspect of model internals and an external condition that the model leverages in carrying out some task. Early work in image classification models suggested that individual units (neurons) within a network may encode visual attributes, such as *red*, or *vertical edge* [Zeiler and Fergus, 2014]. But as models became larger and network architectures more complex, this straightforward mapping between neuron and feature has been questioned, with many MI researchers now hypothesising that features are encoded “in superposition” across multiple neurons—e.g. as almost-orthogonal directions in activation space [Elhage et al., 2022].

To explain neural networks in terms of features is, in philosophical jargon, to adopt a *representational* lens. Representations, in the philosophical sense, are system-internal components whose function is to encode information (or “carry content”) about things external to the system, and thus to drive appropriate behaviour. Feature-based explanations are representational explanations, because they are attempts to explain system behaviour in terms of internal representations, in this sense.⁹ The nature of representations—especially in the context of human and animal cognition—has long been discussed in philosophy, and this remains an active area of research. (For helpful introductions and book-length treatments, see [Ryder, 2009a,b, Shea, 2018, Schulte, 2023]). Here we highlight three ways the philosophy of representation can contribute to MI research into features.

Distinguishing vehicles from content The term “feature” is often used in inconsistent ways, sometimes referring to an internal component of a model—such as a neuron, or a non-basis direction in activation space—and sometimes referring to a property of an input—such as a *curve*, the *Golden Gate Bridge* or *positive sentiment*. Williams [2024] argues that this confusion rests on an equivocation between two aspects of a representation, which it has become standard practice to distinguish in the philosophical literature, namely the *vehicle* and the *content* of a representation. A representational vehicle is the internal symbol, signal, or aspect of activity whose function is to encode content (to detect, represent, or refer to something external). Representational vehicles enter into causal-

⁶Thus, MI practitioners are on firm footing in dismissing the charge that their field is “reductionist” [Hendrycks and Hiscott, 2025].

⁷Recent developments in MI echo this pluralist perspective. The *causal abstraction* framework [Geiger et al., 2024, 2021, 2025] formalises the idea that high-level and low-level descriptions of neural networks can serve different explanatory goals. It is perhaps no coincidence that this work is partly influenced by formal philosophical work on causation and explanation.

⁸Another valuable resource from the philosophy of science can be found in attempts to abstract, systematise, and codify mechanism-discovery strategies often implicit in scientific practices [Darden, 2006, Craver and Darden, 2013, Darden, 2017]. These accounts can help MI researchers to better understand their own toolkits and perhaps inspire new methods through analogies with other fields.

⁹In the ML literature, “representation” is sometimes used in a looser sense to refer to any intermediate activation pattern (irrespective of whether it has a clearly defined content-encoding role). Presumably, though, the term representation is used precisely because they are assumed to play a representational function in the narrower sense (i.e., exploitable encoding of information). We will adopt this narrower usage in what follows.

computational relations with other representational vehicles and ultimately generate the behavioural output of a system. By contrast, the content of a representation is the task-relevant external condition (object, property, category, relation, proposition) that is represented by a representational vehicle, and which makes sense of the cascade of causal interactions between vehicles.¹⁰

One benefit of clearly differentiating the vehicle sense of “feature” from the content sense of “feature” (beyond avoiding unnecessary confusion and cross-talk) is that it allows researchers to clearly articulate distinct research questions. For example, one line of inquiry is *what contents do models learn to represent?* For instance, do language models represent causal information about real-world entities, or do they simply represent syntactic and distributional properties of words? Does a given model represent the states and properties of users? If so, *which* properties? And so on. A quite different line of inquiry is *what are the vehicles of content in ML models?* For instance, when do models develop representational vehicles that align with non-basis directions vs. with individual neurons? How do simple representational vehicles (e.g., those representing objects and properties) combine into complex representational vehicles (e.g., those representing facts)? And what are the *specific* vehicles for *specific* contents—e.g., the vehicle responsible for encoding the language of an input?

Refining experimental approaches A key concern in philosophical discussions of representation has been the search for the right *theory of content*. This is the question of what *grounds* representational content—what makes it the case that a given representation represents *cats* rather than *dogs* or *cat-shaped things*? In the context of AI models, we might similarly ask, what properties of a representational vehicle (e.g. a vector, or a direction in model activation space) determine what it represents? Is content a function of a vector’s relationship to environmental inputs, behavioural outputs, training examples, and/or to other representational vehicles? Here, the philosophical literature can be a rich resource for MI practitioners keen to improve their methods for representational content ascription, i.e. precisely and accurately identifying which contents are represented by which aspects of model internals. For example, Harding [2023] operationalises key criteria from philosophical theories of content, in order to guide MI researchers in testing hypotheses about model-represented contents. Her paper makes concrete recommendations for selecting probes, choosing appropriate causal interventions, and inferring representational contents from these methods. For instance, she argues that the method of Iterative Null Space Projection [Ravfogel et al., 2020] fails to distinguish between hypothesised and confounding representational content assignments. Philosophical theories can also suggest alternative methods for identifying representational contents, which depart from the information-based approaches discussed by Harding. For example, an alternative family of theories appeals to structural correspondences or morphisms between internal activity and external domains [Cummins, 1996, O’Brien and Opie, 2004, Shea, 2014], and philosophers have explored the application of such theories to identify representational contents in LLMs [Søgaard, 2023, Williams, 2026] and earlier neural networks [O’Brien and Opie, 2006, Churchland, 1998].

Suggesting new lines of investigation Finally, philosophical research on representation can suggest new avenues for MI research. As Chalmers [2025] points out, generic talk of “features” tends to obscure the fact that representations in neural networks could in principle represent many different kinds of content: They may represent objects, properties, or relations (concepts, loosely speaking). But they could also represent entire facts or propositions (e.g., that *Paris is the capital of France*). This suggests an open question for MI research — (how) do models construct propositional representations out of sub-propositional representations? Philosophers also distinguish between a proposition and the attitude of a system towards it—*believing* the proposition *the house is on fire* is quite different from *intending* to bring that proposition about. Whether and how such distinctions are realised in the mechanisms of advanced AI systems are important, but under-investigated questions.

2.3 How can we detect deceptive behaviour from model internals?

One major hope for MI is that it can help to detect unsafe or misaligned model processing that is not evident from outward behaviour [Amodei, 2025]. Here, too, philosophy can make important contributions to MI research. While there are many forms of unsafe or misaligned model processing,

¹⁰Philosophers have contributed to discussions over how best to individuate representational vehicles in neural networks for some time [Clark, 1993, Shea, 2007, Azhar, 2016], in many ways prefiguring recent debates about the linear representation hypothesis [Park et al., 2023] in mechanistic interpretability.

we will illustrate our point by focusing on one of the most discussed clusters of issues—deception [Hubinger et al., 2019, Park et al., 2024] and lying [Azaria and Mitchell, 2023, Pacchiardi et al., 2024]. MI researchers hope to detect, anticipate, and mitigate deceptive behaviour by AI systems. As Sharkey et al. [2025] argue, “By monitoring internal representations, [MI methods] could aid in detecting potential sabotage or deceptive behaviour before deployment” (p. 26), with methods such as linear probing already being investigated [Goldowsky-Dill et al., 2025]. However, the very concept of AI deception raises significant philosophical puzzles [Dung, 2025].

Clarifying deception, lying, and related concepts While the concepts of deception and lying are easy enough to grasp intuitively, characterising them precisely turns out to be a challenge. But precise, operationalisable and actionable definitions are needed for MI researchers to target the right phenomena. There is a rich literature in ethics and philosophy of language that attempts to clarify the notions of lying and deception, and distinguish them from neighbouring concepts. A standard view of *deception* in philosophy is that it is the act of intentionally causing another agent to form a false belief [Mahon, 2016]. There are two key elements to this definition: (i) the inducement of false belief in another, and (ii) the presence of an intention or goal on the part of the deceiver [Carson, 2010, Martin, 2009]. *Lying* is a related, but distinct concept. Lying is typically taken to involve making a false statement or assertion to someone, where the speaker does not believe the stated claim to be true [Mahon, 2016]. To lie thus requires the speaker makes a specific assertoric commitment to a false proposition, distinguishing it from other speech acts like jokes or questions [Marsili, 2021]. *Deception* doesn’t always involve lying—one can deceive with one’s actions (e.g., “dummies” and “feints” in sports) or by omitting certain information in conversation without uttering a falsehood.

One payoff of this philosophical literature for MI is to highlight various ways in which one can make false statements or induce false beliefs without lying or deceiving. For instance, in the cases of jokes, metaphorical statements, fiction-writing, and role play, or mere error, one can utter a falsehood that is not a lie, and in the case of accidentally misleading, one can induce a false belief without deceiving.

Moreover, the literature suggests that lying and deception, as traditionally understood, require significant cognitive complexity. More specifically, they require *intentions* on the part of the deceiver and (in the case of lying) the possession of *beliefs*, and the capacity to make *assertions*. These criteria raise an immediate difficulty when applied to AI systems, as it is highly controversial whether even frontier models possess beliefs, goals, and intentions—or even the ability to make statements or assertions—in the relevant sense. Recent philosophical engagement with these latter questions thus offers yet another resource to MI researchers concerned with deception and lying in AI models.¹¹

One tempting move is to weaken the definition of lying or deception, to allow that AI systems could lie or deceive without possessing the psychological states or communicative capacities demanded by traditional accounts. For example, if a model persistently outputs misleading answers in contexts where users predictably misinterpret them, we might label that as deception, regardless of the model’s internal “motivations” [Tarsney, 2025]. However, the idea that insights from MI can help to develop methods for detecting deceptive behaviour based on model internals is premised on a richer notion of lying and deception, one on which the internal states of a system, and not just its behaviour or its effect on users, are relevant. Another approach is to attempt to identify states like beliefs, intentions, and speech acts—or close functional analogues of them—in AI systems. In the next section, we discuss such attempts in MI research and show how philosophy can play a guiding role.

Guiding the search for AI “beliefs” A key strategy for lie-detection via MI has been to attempt to identify language models’ “beliefs” from internal states and to detect mismatches between beliefs and outputs. Initial studies used probing classifiers to identify directions in models’ internal activations that correspond to the truth value of inputs. For instance, Azaria and Mitchell [2023] and Burns et al. [2023], using different probing methods, presented evidence that the truth value of inputs could indeed be decoded from model activations. Azaria and Mitchell frame this project as “extract[ing] the LLM’s internal belief” (2023, p. 2) and gave their paper the bold title “The Internal State of an LLM Knows When It’s Lying”. Should we accept these claims at face value? Belief is a central concept in philosophy, particularly in the sub-fields of philosophy of mind, epistemology, and philosophy of action, so this is a natural place where philosophical engagement can add value to MI research.

¹¹See e.g. [Butlin and Viebahn, 2025, Williams and Bayne, 2024, Skiba, 2026] on assertion and [Williams, 2025] on intentions. The question of beliefs in AI models is discussed further in the text below.

The philosophers Ben Levinstein and Daniel Herrmann [Levinstein and Herrmann, 2024, Herrmann and Levinstein, 2025] have raised questions about the face-value interpretation of the above findings. For instance, they identified generalisation issues with existing probes, and also argued that for an information-carrying state to qualify as a belief, it must be shown to be used by the system—i.e., causally drive behaviour appropriate to the content of that belief (see also Harding 2023). This critique has directly informed subsequent MI research: Marks and Tegmark [2024] curated new datasets to address generalisation issues identified by Levinstein and Herrmann and took steps to establishing through interventions that these components causally mediate outputs appropriately.

In a follow-up paper, Herrmann and Levinstein [2025] expand on their previous conceptual work by proposing four criteria for LLM representations to count as beliefs, grounded in existing philosophical literature on belief. They also offer some meta-reflections on the nature and utility of concepts like “belief”, suggesting that rather than being a binary issue, “[t]he satisfaction of these requirements come[s] in degrees; in general, the more a representation satisfies these requirements, the more helpful it is to think of the representation as belief-like” (p. 7). This dialogue illustrates yet another domain in which philosophy can contribute to MI research. By clarifying concepts like *belief*, philosophers can help refine the methods by which MI researchers identify belief-like representations in AI systems, and thus ultimately improve “lie-detection” methods.

Drawing ethical and normative boundaries MI techniques may be able to identify circuits that are deceptive or suppress information during evaluations. However, not all cases of deception are ethically equivalent. As a result, developers face complex questions about which to intervene on and which to preserve. These decisions require normative frameworks that current MI research lacks.

Consider three hypothetical “deceptive” circuits that may be discovered using MI. The first detects regulatory oversight procedures and triggers the concealment of malicious capabilities (e.g., capacity for producing malware). The second suppresses the home addresses of public figures. The third withholds diagnostic information for certain medical prompts (e.g., “sharp chest pain and difficulty breathing”) and redirects users to seek out emergency medical care. Standard MI approaches might flag all three mechanisms as instances of the same “deceptive” mechanism since they all conceal information contained in the model. Yet, each circuit has distinct ethical implications requiring different interventions. In particular, the first circuit selectively conceals capabilities to circumvent regulatory constraints and should reasonably be targeted for removal. The second circuit protects information that users have no legitimate right to access, and should likely be preserved and potentially strengthened. Finally, the third circuit redirects users to urgent medical care for their own safety and should therefore be preserved and constantly updated to ensure its medical urgency detection is accurate while preserving the core redirection function.

Integrating ethics into MI methodology can provide ethical normative frameworks that assist developers in distinguishing ethically problematic cases of information suppression and deception that demand intervention from those that are ethically benign [see Danaher, 2022, Sætra, 2021, Kneer, 2021]. For example, philosophical work on manipulation, nudging, and epistemic agency can help researchers reason about which kinds of model behaviour warrant intervention or risk mitigation and which do not [Barnhill, 2022, Pepp et al., 2022]. Careful ethical analysis is crucial to avoid downstream harms of broad-brush approaches to detecting and dealing with deception.

3 Objections and responses

The proposal that philosophy has a central role to play in mechanistic interpretability (MI) may be met with skepticism. We address six common objections below and argue why they do not undermine the value of sustained philosophical engagement in the field.

“Armchair theorising won’t get us anywhere.” A common worry is that philosophical contributions are overly abstract or speculative. These “armchair” exercises are disconnected from empirical reality and risk missing the complexities of real-world interpretability problems. We think that this is a very valid worry for some philosophical approaches that emphasise *a priori* reasoning and “in-principle” arguments. However, much of contemporary philosophy, and especially MI-relevant fields such as philosophy of science, are deeply engaged with empirical research. Many of the philosophical works cited in this paper are informed by close engagement with bleeding-edge empirical findings. Some philosophers even get their hands dirty attempting replications of MI studies

[Levinstein and Herrmann, 2024, Lederman and Mahowald, 2026] and operationalising concepts for empirical investigation [Harding, 2023]. Armchair philosophy of mechanistic interpretability may be a non-starter, but philosophy need not be practised from the armchair.

“MI researchers can do the philosophising themselves.” A second potential objection is that the philosophical dimensions of MI can just as well be addressed internally by MI researchers without needing philosophers. We grant that scientists can contribute to answering philosophical questions about their field, especially those related to the foundations, methods, and implications of scientific practice.¹² But while many MI researchers do engage with philosophical questions (as exemplified by Sharkey et al. [2025]), doing so effectively requires philosophical training and knowledge of philosophical literature. As illustrated in this paper, philosophy has many tools, distinctions, and debates that are unfamiliar or underutilised in ML. And crucially, because philosophers are not burdened with technical implementation, they are often better placed to “see the forest for the trees”, question framing assumptions, and ask big-picture questions [Bickle et al., 2024]. Collaboration between MI researchers and philosophers make the most of each party’s expertise, and can mitigate the ever-present risk of reinventing the wheel.

“Philosophers aren’t well informed about MI.” To be clear, our claim that mechanistic interpretability needs philosophy should not be read as an injunction aimed solely at MI researchers. Realising the potential contribution of philosophy in MI will equally require more philosophers to direct their efforts and attention to MI, and develop the technical literacy to do so usefully. That said, while many philosophers currently lack the relevant expertise, there is a growing cohort with interdisciplinary training, including backgrounds in computer science, neuroscience, or cognitive modelling. Many philosophy of science programs require coursework in both philosophy and specific scientific disciplines, ensuring that students gain technical foundations alongside philosophical training. And communities of empirically-oriented, technically-literate philosophers exist within most branches of philosophy, including philosophy of mind, epistemology, ethics and philosophy of language.¹³ Moreover, as noted above, the key to productive collaboration is not perfect symmetry of expertise, but mutual recognition of complementary strengths.

“Philosophy can foster progress in MI? Prove it!” The claim that philosophy has the potential to catalyse progress in mechanistic interpretability may be met with suspicion, absent a proven track record. But a number of prominent papers in the field have engaged profitably with philosophical work. For instance, influential MI work on causal abstraction and causal mediation analysis, such as [Geiger et al., 2022, 2024, 2025] and [Mueller, 2024, Mueller et al., 2026], cites philosophical work on causation and explanation [Hume, 1748, Pearl, 2000, 2001, 2009, Salmon, 1984, Lewis, 1986, 2000, Woodward, 2005, 2021, Halpern, 2016a,b, Yablo, 1992, Creel, 2020]. And recent MI work at Anthropic on introspection in language models [Lindsey, 2025] discusses philosophical work on introspection [Long, 2023, Song et al., 2025, Kammerer and Frankish, 2023] and consciousness [Rosenthal, 2005, Carruthers, 2017, Searle, 1992, Chalmers, 1995, Block, 1995].

“MI researchers already engage with philosophy.” One might turn the previous objection on its head: MI already cites philosophy, so there’s no problem to be discussed here. However, the handful of MI papers just discussed represent exceptions, not the rule. While philosophers are increasingly engaging with MI research, MI engagement with philosophy remains a minority trend. As we’ve illustrated in this paper, there are countless avenues for philosophical contributions to MI that remain unexplored, and philosophical insights have yet to be absorbed into the field at large. Thus, we contend that philosophy’s full impact on the field of MI has yet to be realised.

“What is the practical suggestion here?” The claim that MI needs philosophy might seem intractably vague. We close with some concrete recommendations, in increasing order of institutional ambition. *At the level of individual practice*, MI researchers can read more philosophy;¹⁴ attend philosophy workshops and seminars; convene interdisciplinary reading groups; and seek philosophical

¹²Alan Turing is an early and notable example (see Turing [1950]) within the field of computer science.

¹³This paper has discussed many examples of technically-literate philosophical work that engages with MI research. For additional examples see [Beckmann, 2025, Beckmann and Queloz, 2026, Budding, 2025, Coelho Mollo and Millière, 2026, Goldstein and Levinstein, 2024, Grindrod, 2026, Keeling and Street, 2025, Millière and Buckner, 2024, Yetman, forthcoming].

¹⁴For authoritative and up-to-date overviews of philosophical topic areas, we recommend the Stanford Encyclopedia of Philosophy; the journal *Philosophy Compass*; Oxford University Press and Routledge’s “Handbook”, “Companion” and

feedback on manuscript drafts. Philosophers, reciprocally, can engage seriously with MI papers (and develop the technical literacy to do so); follow the MI literature; and write for venues that MI researchers actually read. *At the level of research groups and institutions*, we recommend co-supervision of graduate students across departments; embedding philosophers-in-residence at MI labs (and reciprocal visits by MI researchers to philosophy departments); summer schools and fellowships designed explicitly for cross-training; co-authored empirical-conceptual projects; and grant calls that require interdisciplinary teams. And *at the level of the field*, we encourage dedicated philosophy workshops and symposia at machine learning conferences with reciprocal MI sessions at philosophy meetings; position-paper tracks in machine learning venues (like the one hosting this paper), with editorial and reviewer expertise in philosophy; interdisciplinary journals; and undergraduate and graduate curricula that train researchers in both technical and philosophical methods.

4 Conclusion

Mechanistic interpretability is a rapidly evolving field, driven by urgent practical needs and rich with conceptual complexity. As we have argued, philosophy is deeply relevant to this field. MI raises foundational questions about explanation, representation, knowledge, agency, and values, and meaningful progress in the field is often predicated on handling these concepts carefully. As illustrated above, philosophers can help provide conceptual clarity, identifying and scrutinizing assumptions, proposing novel research questions, interpreting empirical results, and illuminating ethical complexities. To stress, our claim that mechanistic interpretability needs philosophy is neither a call for disciplinary silos to remain intact, nor a claim that philosophy has ready-made answers for all of the challenges encountered in MI research. Instead, it is an invitation to deeper interdisciplinary collaboration that is technically informed and philosophically grounded. As AI systems become more powerful and more deeply embedded in society, the stakes of understanding them, not just how they behave, but how they work, have never been higher. Enriched by philosophy, mechanistic interpretability has a clearer shot of success.

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